Dataset Quality Criteria for UAV-Based Structural Defect Detection

1 Introduction

To ensure reliable training and evaluation of bridge defect detection models using UAV-based imagery, it is important to define structured dataset quality criteria. These criteria help in filtering low-quality data and improving annotation quality. This document defines three key criteria: Image Quality, Defect Visibility, and Annotation Quality.

2 Image Quality

Image quality refers to the visual clarity and usability of an image for automated defect detection. It is determined using the following measurable components, but I need to search for their threshold scientifically, so that we can filter the best quality images. I have already used the plugin to add the blur score, brightness score, and contrast score in FiftyOne.

- **Blur Score:** Measures image sharpness. Low sharpness indicates motion blur or defocus, which makes the defects difficult to see.
- **Brightness Score:** Evaluates exposure levels. Underexposed (dark) or overexposed (bright) images reduce the visibility of defects such as cracks and corrosion.
- **Contrast:** Measures the intensity difference between foreground and background pixels. Low contrast images make it difficult to distinguish defect details.
- **Resolution:** Represents the spatial detail of the image (width $\times$ height). Low-resolution images may fail to capture small scale defects.

3 Defect Visibility

Defect visibility will describe how clearly structural defects can be observed in an image from a human perspective. This criterion is manually assessed and categorized as follows:

- **Highly Visible:** Defects are clearly observable and easily identifiable(sharp edges)
- **Low Visible:** Defects are present but difficult to observe due to far distance, blur, occlusion, or low resolution.

This criterion is especially important in UAV-based datasets, where altitude and viewing distance significantly affect defect perception.

4 Annotation Quality

Annotation quality refers to how correct, consistent, and complete the labels are. It ensures that training models are built on accurate ground truth information. It is defined through the following components:

- **Accuracy:** Are the labels actually correct?
- **Completeness:** Are all defects annotated?
- **Consistency:** Are all annotations done in the same way? Do the annotations follow the guidelines?